



AI.1 project

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AI Traffic Control System

Manar Samir Abdulraheem

Mennallah Gamal Mohamed Mohamed

Menna Allah Mohamed Magdy

Section 8

Supervised by:

DR/ Sara EL-Metwally

E/ Asmaa Nagib

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Problem description

Title: AI Traffic Control System.

Description:

The Intelligent Traffic Light Control System is an AI-based simulation designed to optimize traffic flow at a four-way intersection. Traditional traffic light systems operate on fixed timing schedules that do not adapt to real-time traffic conditions, which can lead to congestion, increased waiting time, and inefficient traffic management. The main objective of this system is to minimize average waiting time, maximize traffic throughput, and ensure fairness between directions.

PEAS

Performance Measure:

1. Minimize average waiting time.
2. Maximize number of cars passing.
3. Ensure fair distribution of green/red signals.
4. Reduce traffic congestion.

Environment:

1. Four-way traffic intersection.
2. Vehicle arriving randomly.
3. Traffic lights controlling directions.
4. Multiple vehicle queues.

Actuators:

1. Traffic lights (Red/Yellow/Green).
2. Change signal timing.
3. Select direction with green signal.

Sensors:

1. Number of cars in each direction.
2. Waiting time of vehicles.
3. Current signal state.
4. Traffic queue length.

ODESDA

1. **Observable** => as we can see the complete state of the whole city and make decision about the traffic manage
2. **Stochastic** => Traffic flow is affected by random events such as unpredictable vehicle arrivals, accidents, weather conditions, or road closures, making the environment partially deterministic.
3. **Sequential** => The environment is sequential because decisions made by the agent (such as adjusting signal timing at intersections) affect future traffic flow across the city.
4. **Dynamic** => The traffic environment continuously changes due to moving vehicles, new arrivals, and varying traffic patterns throughout the city.
5. **Single agent** => A single centralized traffic management agent controls and coordinates traffic signals across the entire city to optimize traffic flow.
6. **Continues** => as the flow of the car is infinite

Agent type

- Utility-based agent.